



Complete Guide to Using the DKC-Pro Color Chart

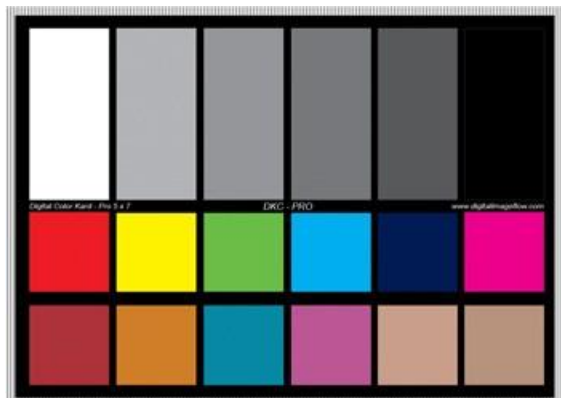
Congratulations on your purchase of the DKC-Pro Color Chart. With tens of thousands sold worldwide, DGK™ is a leading manufacturer of precision color balance tools in the USA and Europe. DGK products are chosen by Nikonians™ digital photography workshop instructors, Adobe Photoshop™ instructors, and professional photographers everywhere. DGK color tools are used in medical imaging systems, forensic photography, underwater photography and video, computer and robotic vision systems, and commercial production applications. With proper use of your DGK product, you can be assured of *perfect digital color every time™*.

Visit the FAQ page on our website, dgkcolor.tools, for answers to many common questions on the various features of all the DGK color tools. You can also download the Quick-Start User's Guide in German, French, Italian, Spanish, Russian, Chinese, Hebrew, Arabic, and English.

1.0 What is a DKC-Pro Color Chart?

The DKC-Pro Color Chart is a 5" x 7" color reference chart (chip chart for filmmakers) with 18 color patches on one side and a large 18% gray color patch on the reverse. The DKC-Pro is a white balance, exposure reference, and color reference tool for film, digital and video photographers. The product is the result of two years of testing and research in our labs and extensive field testing by photographers, videographers, teachers, colorists, website creators, and others. The color patches are now 100% coated with our exclusive N-Chrome process. The DKC-Pro provides photographers with all the color, white balance, and exposure tools they need in a compact and economical format, with an unprecedented level of color accuracy and color fidelity.

1.1 Features of the DKC-Pro Color Chart



--- Multiple calibrated white balance targets for precise digital white balance and color balance in all lighting conditions.

--- Large 18% gray target for accurate and uniform film and digital exposure under all lighting conditions. Our N-Chrome coated patches are the most accurate available.

-- 6 step grayscale for exposure setting using the histogram function of DSLR's and HD video cameras. Set dynamic range, white levels, black levels, and gamma of HD video and DSLR cameras in video mode.

-- 18 scientifically prepared color targets that represent the primary colors, process colors, and natural colors such as human skin, flowers, sky, fruit, and foliage. These color patches are 100% N-Chrome coated. The card creates an

objective reference you can use in the studio or in your post-production work to compare and analyze any color differences or errors that may have occurred. In this way you can be assured of perfect digital colors from capture to display to print.

--- Creates custom camera profiles when used with color profiling software available on our website and Adobe Camera Raw. Individualized precision camera profiles insure optimum color reproduction for critical applications.

1.2 A Brief Introduction to White Balance in Digital Photography:

White balance (WB) for film photography was determined by the film type and by the film lab. The photographer's role was limited to choosing the appropriate film. Daylight and indoor (tungsten) balanced film were available. In addition, color balancing filters could be placed over the lens. Most WB happened at the lab. The drug store print machine had an automatic color balance process, and the technician could check the prints and correct the colors if needed. As a result, most film photographers were completely unaware of the problem of WB. With digital photography, things changed. In digital photography there is no drug store machine and no technician, so the photographer must manage WB herself. Digital cameras have automatic white balance (AWB), which often works quite well. However, AWB can perform very poorly under certain conditions, even in high-end digital cameras. Once a photographer moves beyond camera phones and simple point and shoot cameras, she will notice that there are problems with color balance in some of her photographs. At that moment, she realizes she needs a simple, portable, and easy to use WB tool.

1.3 Gray Cards for White Balance Setting in Digital Photography:

All digital cameras now have AWB. The AWB function of digital cameras attempts to find a *spectrally neutral* object in each photo, and it uses this to set the WB. All spectrally neutral objects are some shade of neutral gray, anywhere from pure white to very dark or black (Figure 1).

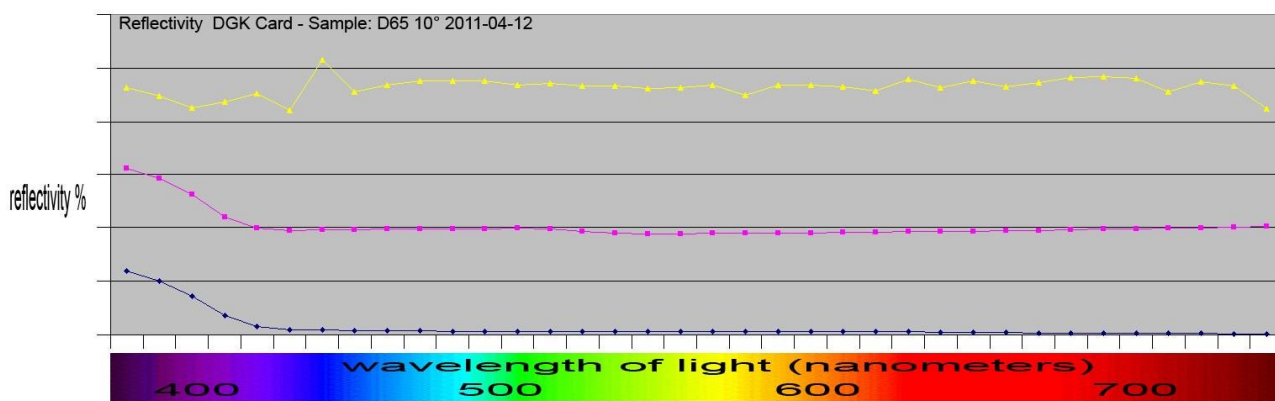
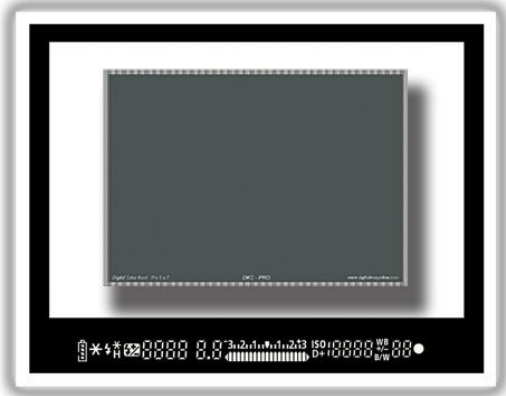


Figure 1: Reflectivity data from the DGK R27 Digital White balance target shows spectral neutrality. The top line shows the white color patch data, the middle line the gray color patch data, and the bottom line is the black color patch data. Spectrally neutral objects may be any shade of gray.

Since not all scenes contain a neutral reference, photographers need to carry one with them, and use it when they want to get the most accurate colors possible. The DKC-Pro has a large spectrally neutral 18% gray patch on the back, and 5 perfectly neutral patches of various lightness on the front (patches 1-5) which are calibrated for ultra-accurate white balance. Each WB patch is scientifically formulated to be spectrally neutral under all lighting conditions. As a result, the light reflected from the DKC-Pro is a true and accurate representation of the physical qualities of the light illuminating the card. Nothing is added, nothing is distorted, and nothing is taken away. To perform a color balance operation, the photographer uses the camera's controls to register the light reflected from the DKC-Pro as a reference. Using this reference, the camera or the post processing software performs a precision balancing of all the colors in the scene.

2.0 Quick- Start Guide for In-Camera Custom White Balance (for JPEG Shooters)



To use the DKC-Pro as the basis for an in-camera custom white balance, you will need to review your camera's manual and become familiar with "custom white balance" operations. Set the camera to the custom white balance setting. Use patch 19, the large gray patch on the back of the card for this function. Place the DKC-Pro so that it is illuminated by the same light as your subject. Make sure enough of the center of the frame is covered by the card in your picture. Consult your camera's manual to see how much of the frame should be filled. Go through the menu to select that picture for your custom white balance. You can save these images and settings and re-use them at any time. Remember, you will need to shoot a new custom white balance shot each time you go into a different room, or the sun goes under a cloud, or if you are pointing your camera at a portion of

the scene that is more strongly lit by one particular light source. If reflections are seen on the DKC-Pro surface during a white balance operation, reposition the card so that reflections are eliminated.

2.1 Quick- Start Guide for Post-Processing White Balance (for RAW Shooters)



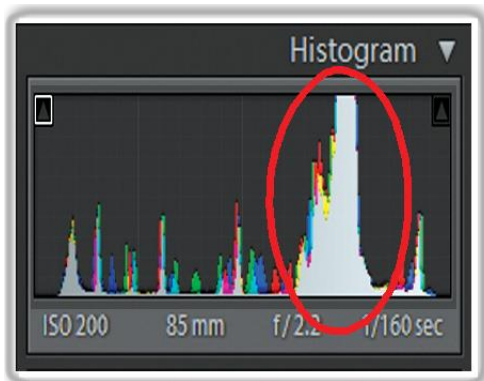
When shooting RAW, patches 2, 3, 4 on the DKC-Pro are recommended. All these patches are spectrally neutral, which patch is best depends on your personal preference and the intensity of the light in your shot. Shoot WB images of the DKC-Pro at the beginning and end of each series of shots. The DKC-Pro does not need to be exactly in focus or centered. When shooting a wedding, for example, walk through the location and shoot the DKC-Pro in each area of the house, church, or reception where you expect to shoot later. Hold the DKC-Pro out in front of yourself and use a wide-angle lens so you can see where you were when you made the shot. Later, find the appropriate DKC-Pro reference shot to use for WB in your RAW image processor.

Select all the images taken under that lighting and simply sync the WB for the whole series. The black and white patches are used to set black and white points in the Photoshop levels control, or to control the output of your photo printer. See the instructions for your printer or image editing software.

2.2 Setting Exposure Using the 18% Gray Patches for Film and Digital:

For exposure setting, place the card in the same light as the subject, and obtain a spot meter reading off patch 4 or 19. Note: Gray cards are used for exposure in all types of photography. This function of a gray card depends on its reflectance (lightness or darkness), and is distinct from the white balance process, which depends on the spectral neutrality of the card. An 18% gray card is necessary for correct exposure in film, digital and video.

In exposure setting, the camera's build-in meter is set to adjust the entire scene to 18% reflectance. Very bright areas, such as glass and mirrors, or large bright areas in the scene may result in the camera setting a very short exposure, which would lead to underexposure of the darker areas of the scene. An 18% gray card is the perfect tool to ensure that the metered light reading is always accurate.



2.3 Advanced Exposure Techniques for RAW Shooters

Due to the linear data process used by digital camera sensors, RAW captures allow the photographer to “expose to the right” and preserve image detail that would otherwise be lost. More on this advanced topic is available at our website. For maximum image detail, increase exposure so that the DKC- Pro grayscale data is pushed 2/3 to 3/4 to the right on your camera histogram. (Figure 3) Your images will appear overexposed in your camera LCD, but processing the RAW images in your software later will eliminate the overexposure and reveal maximum image detail. For best results, always bracket your exposures.

2.4 Using the DKC-Pro as an Objective Color Reference

The 18 color patches include primary colors, process colors, and natural colors such as human skin, sky, flowers, and foliage. These scientifically prepared colors reflect light from the entire visible spectrum of light in a consistent, accurate way. Use the DKC-Pro chart as an objective color reference in the studio or in your post-production work to compare and analyze any color differences or errors that may occur. To evaluate the effect of adjustments to your software or hardware on the accuracy of color display or reproduction, simply compare the image of the chart with the actual chart. See our website for colorimetry data.

2.5 Using a DKC-Pro for Color Balance in Printed Photographs

Place the DKC-Pro in the scene and record one exposure with the card and one without. During the printing process, the DKC-Pro can be used as a reference for setting the brightness and color balance of the printer or of the final print. The white and black target areas can be used to set values using the “levels” control in Adobe Photoshop, Photoshop Lightroom, and other digital image editing software tools.

3.0 Creating Custom Color Profiles Using the DKC-Pro

Please see the accompanying guide and software on our website.

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DKC-Pro Colorimetry Data

1 White 97,0,1	2 Gray 73 73,0,0	3 Gray 62 62,0,0	4 Gray 50 50,0,0	5 Gray 38 38,0,0	6 Black 23,0,0
7 Red 48,59,39	8 Yellow 92,1,95	9 Green 64,-40,54	10 Cyan 57,-41,-42	11 Blue 18,-3,-25	12 Magenta 49,60,-3
13 CIE TSC 01 41,51,26	14 CIE TSC 02 61,29,57	15 CIE TSC 06 52,-24,-24	16 CIE TSC 08 52,47,-14	17 CIE TSC 09 69,14,17	18 CIE TSC 10 64,12,17